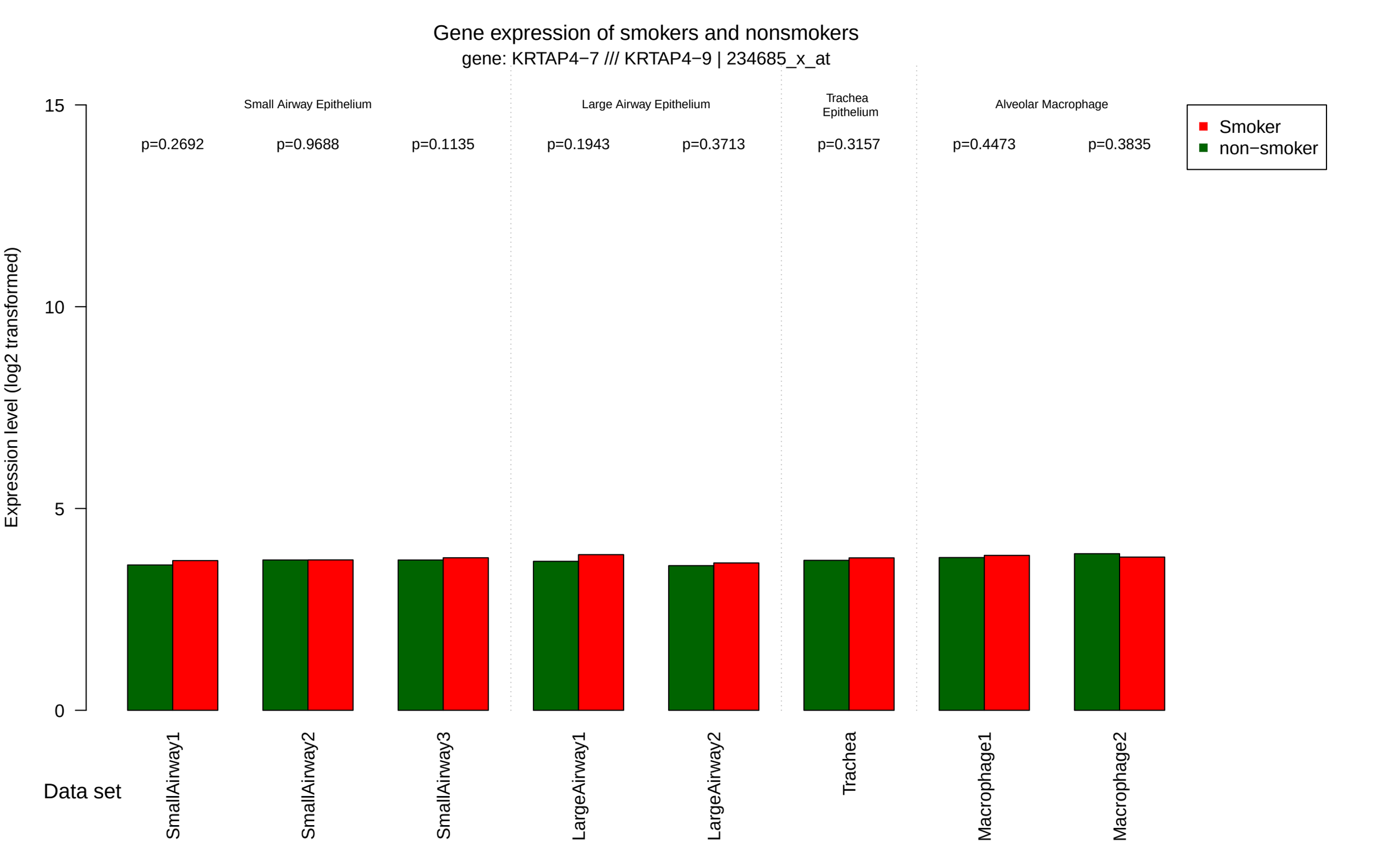
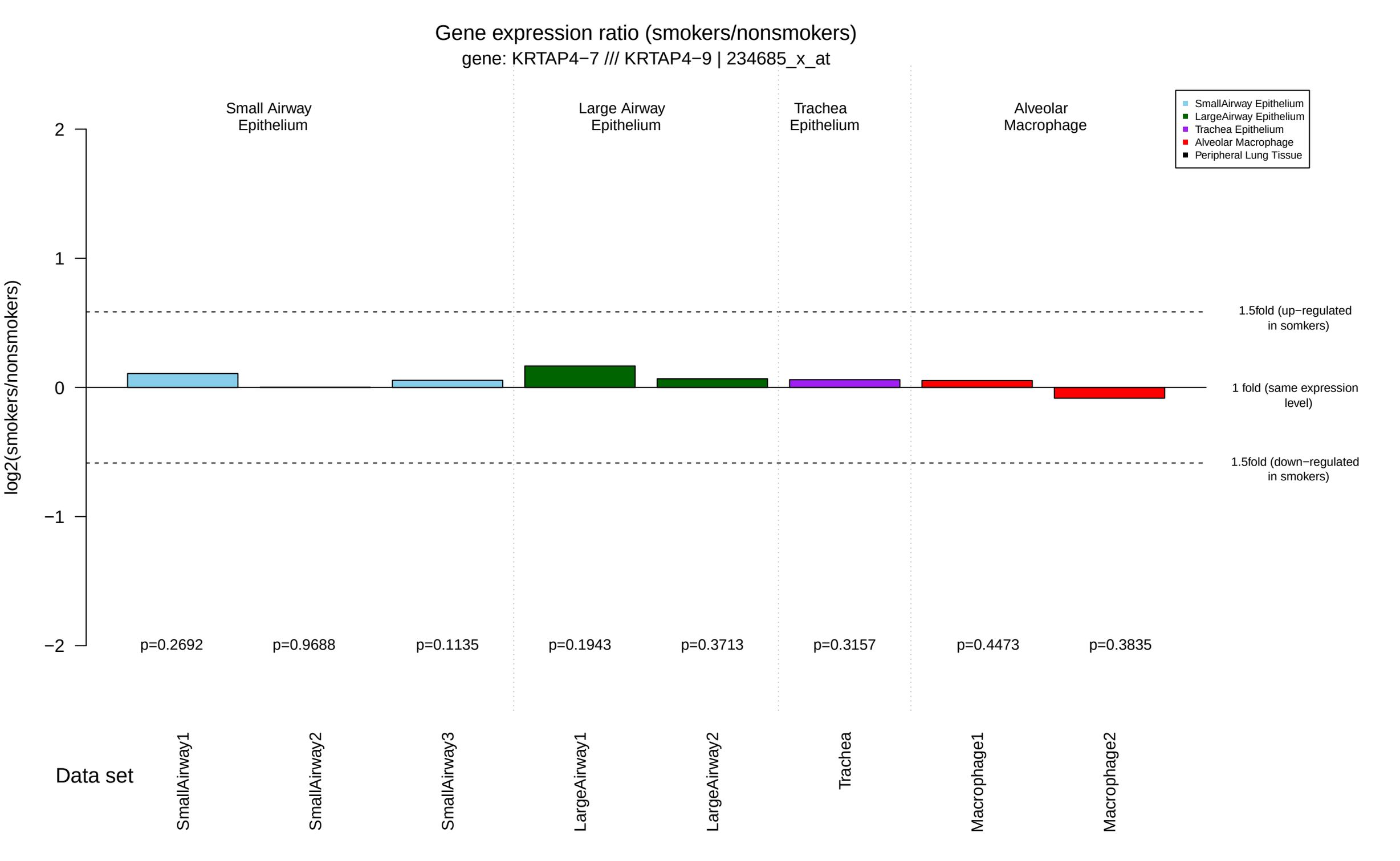
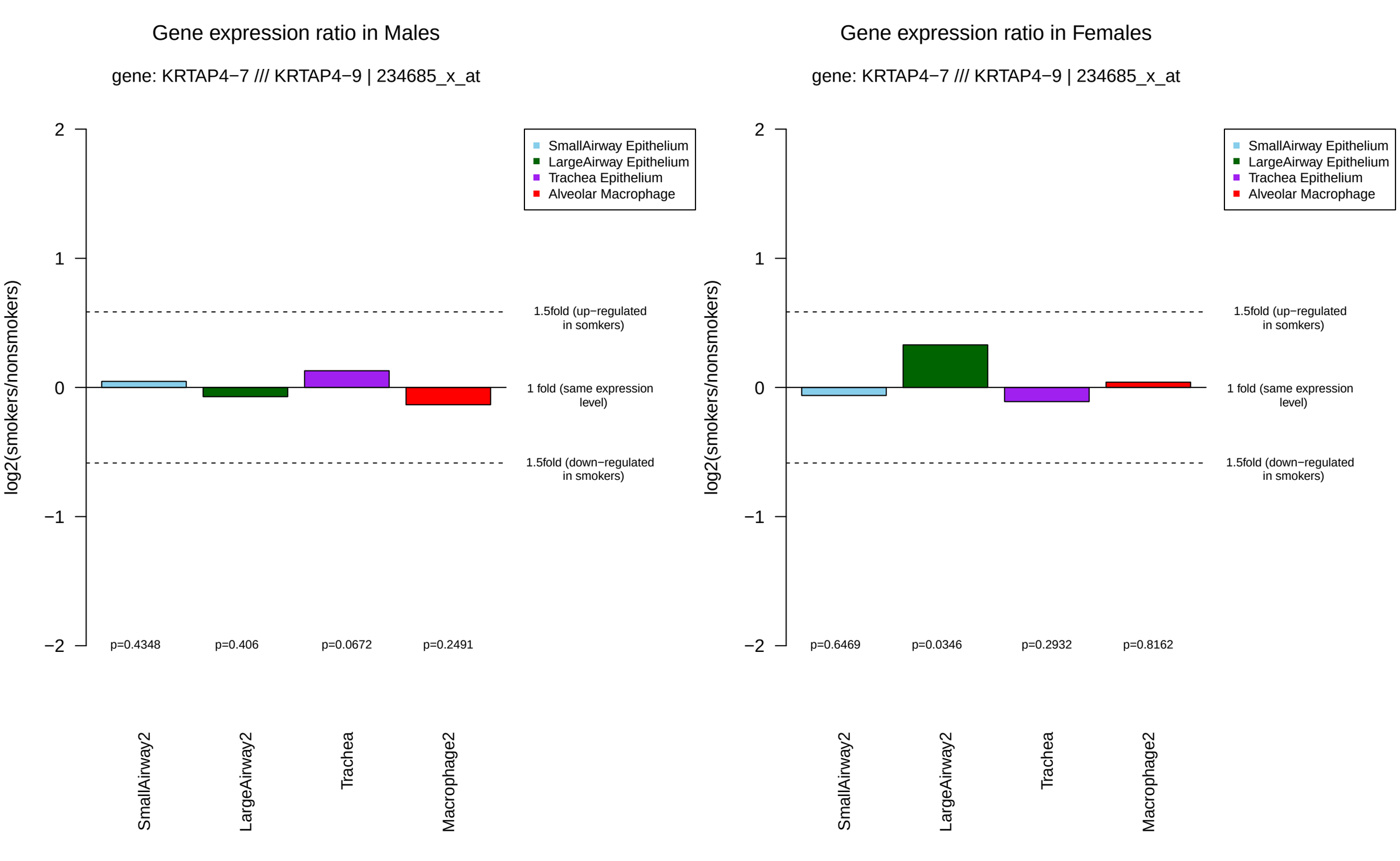
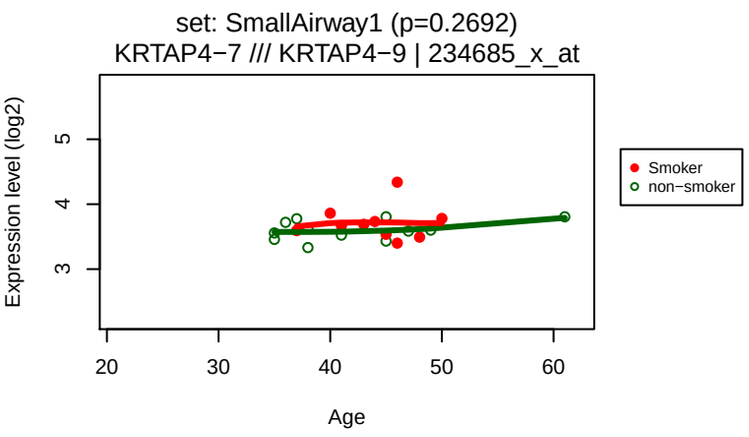
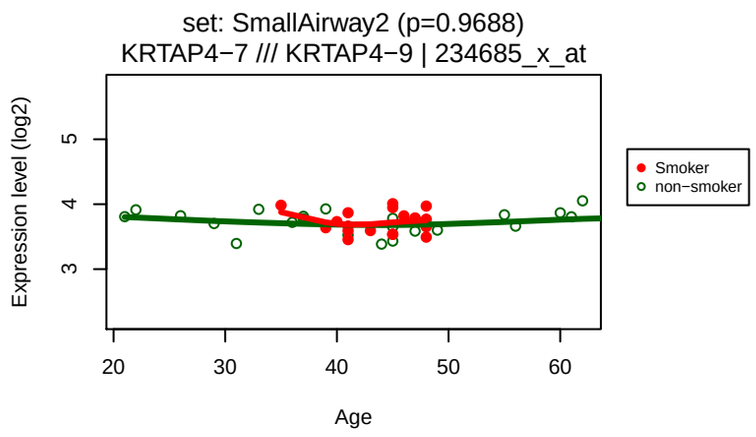
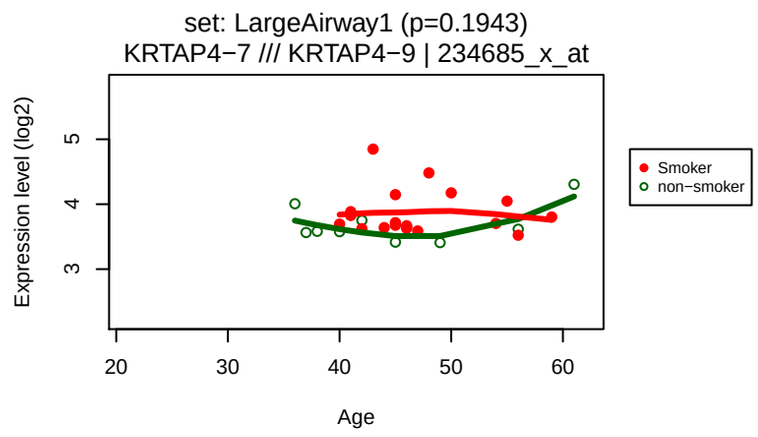

Gene expression of smokers and nonsmokers
gene: KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2692
p=0.9688
p=0.1135
p=0.1943
p=0.3713
p=0.3157
p=0.4473
p=0.3835
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

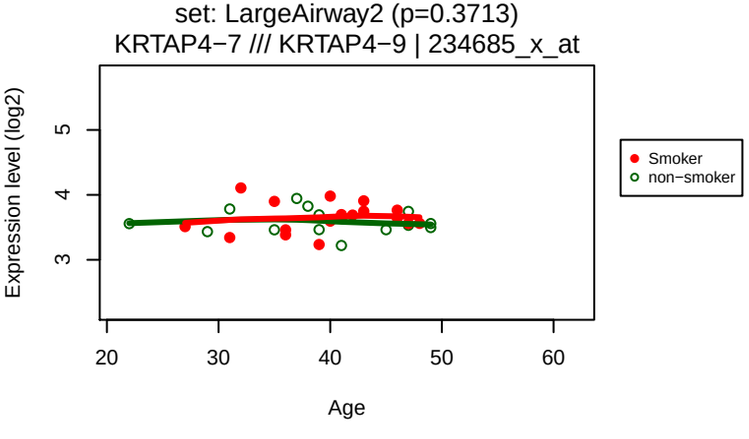

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2692
p=0.9688
p=0.1135
p=0.1943
p=0.3713
p=0.3157
p=0.4473
p=0.3835
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

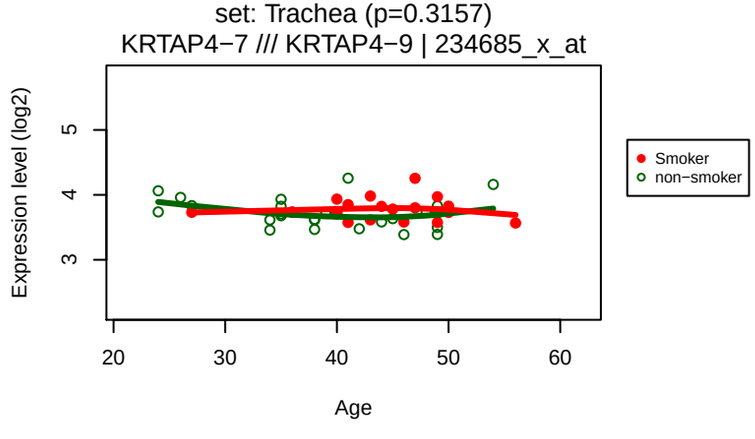

Gene expression ratio in Males
gene: KRTAP4−7 /// KRTAP4−9 | 234685_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4348
p=0.406
p=0.0672
p=0.2491
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP4−7 /// KRTAP4−9 | 234685_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6469
p=0.0346
p=0.2932
p=0.8162
SmallAirway2
LargeAirway2
Trachea
Macrophage2

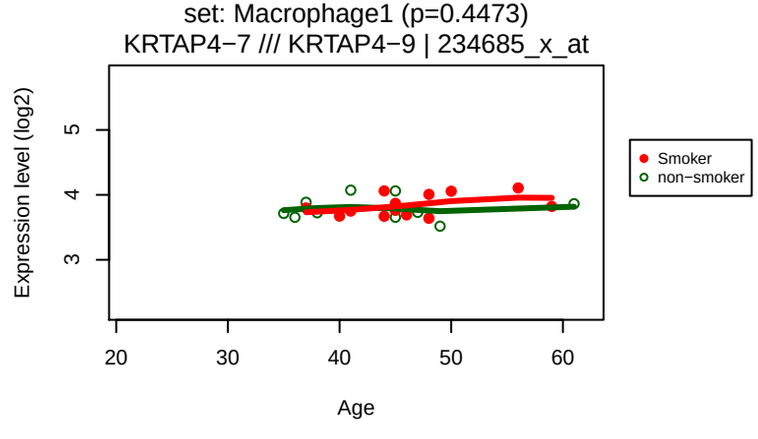

set: SmallAirway1 (p=0.2692)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker

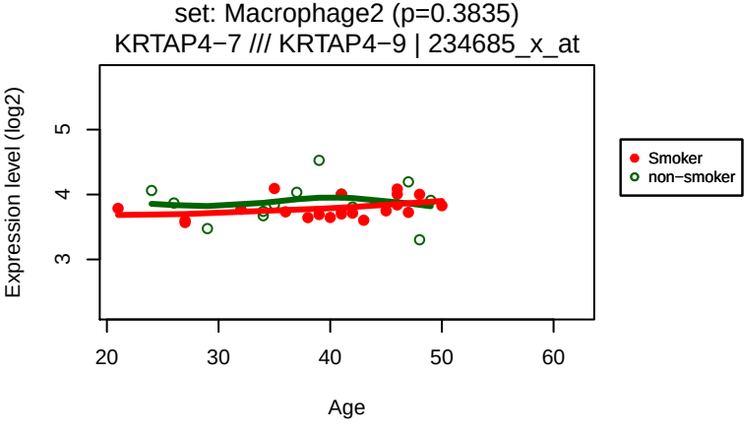

set: SmallAirway2 (p=0.9688)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker

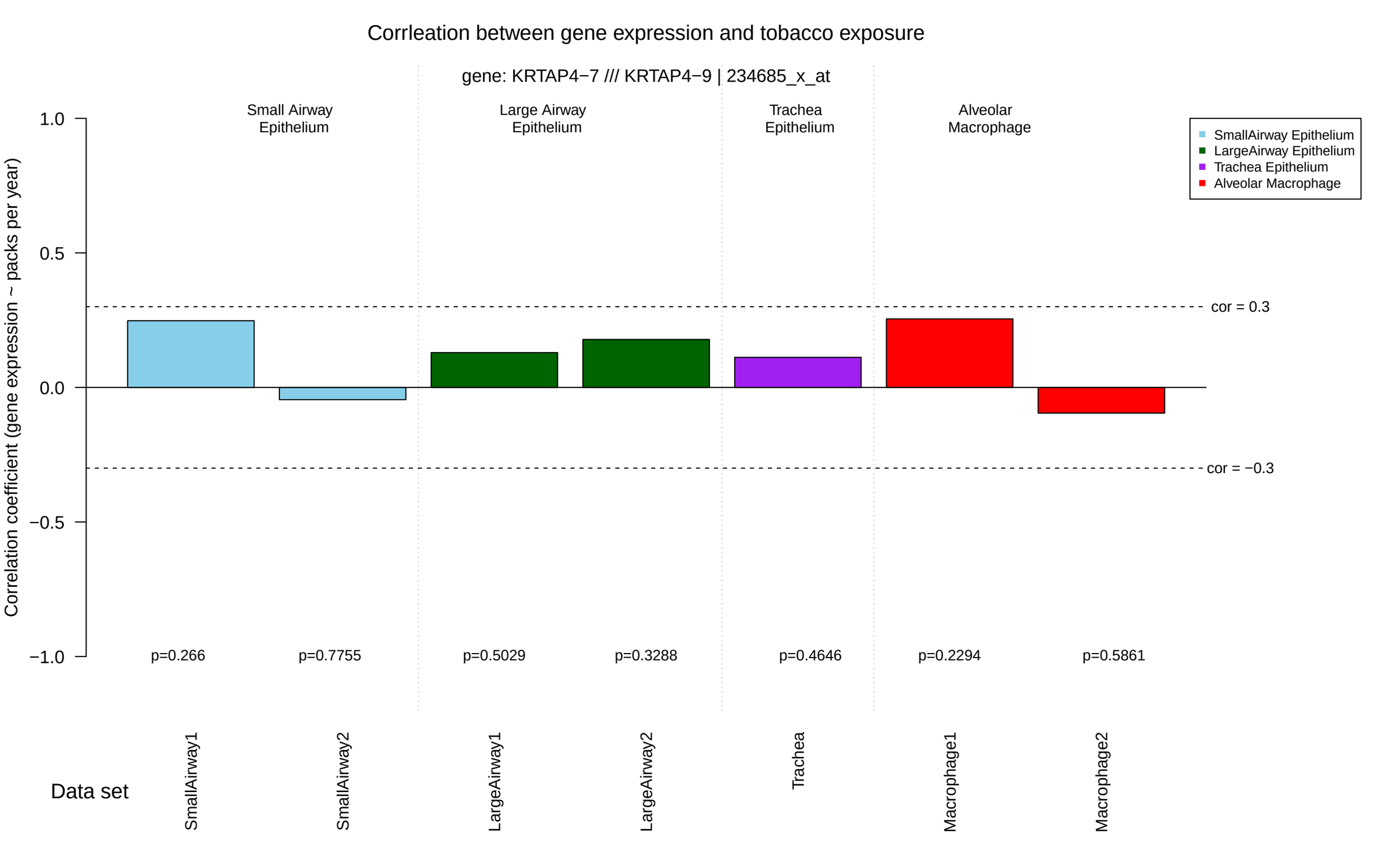

set: LargeAirway1 (p=0.1943)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3713)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3157)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4473)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3835)
KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP4−7 /// KRTAP4−9 | 234685_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.266
p=0.7755
p=0.5029
p=0.3288
p=0.4646
p=0.2294
p=0.5861
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2